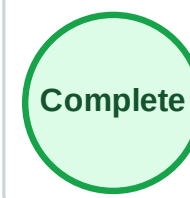


BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing

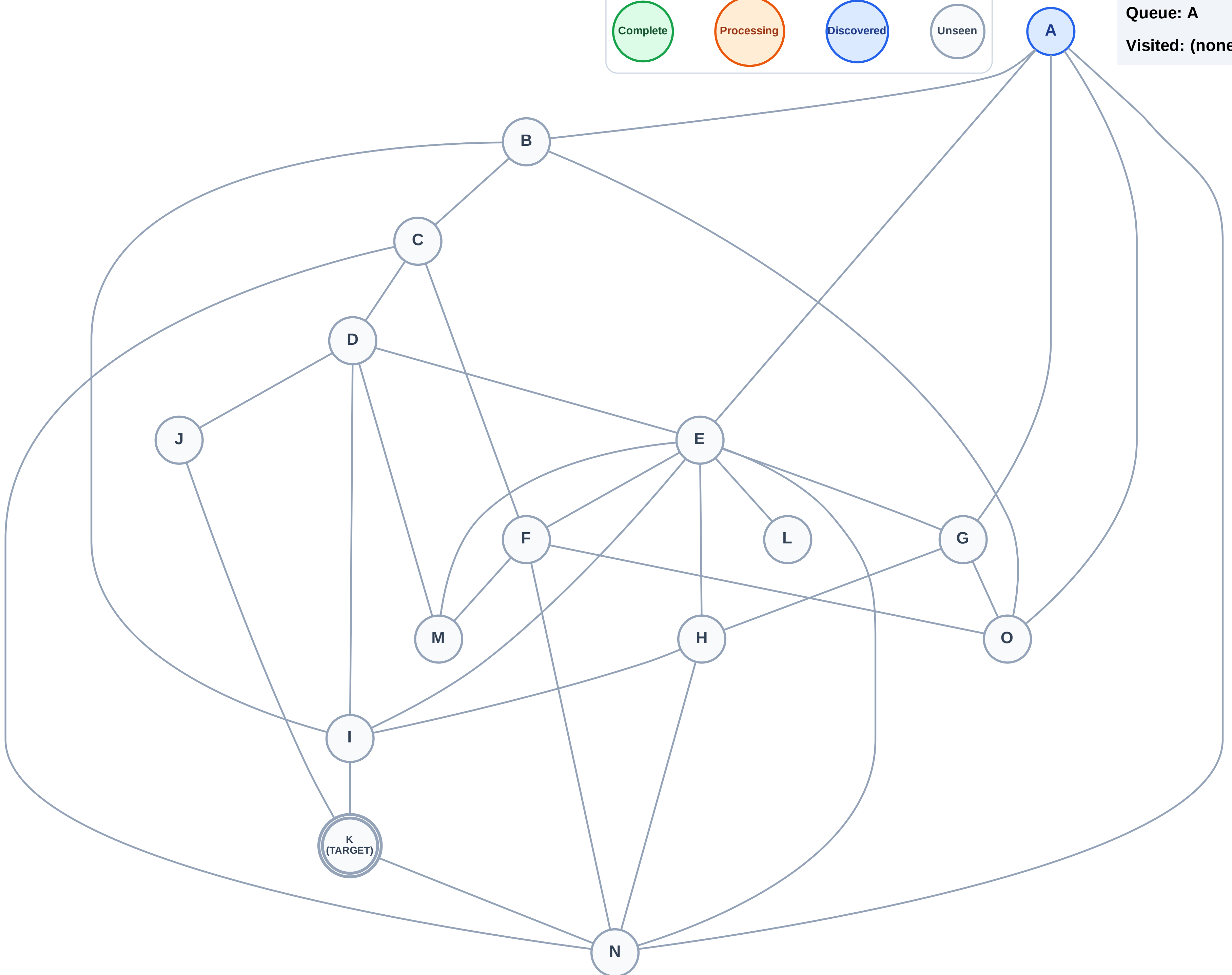


Discovered



Unseen

Queue: A
Visited: (none)



BFS Traversal
Step 2: Process node A

Legend

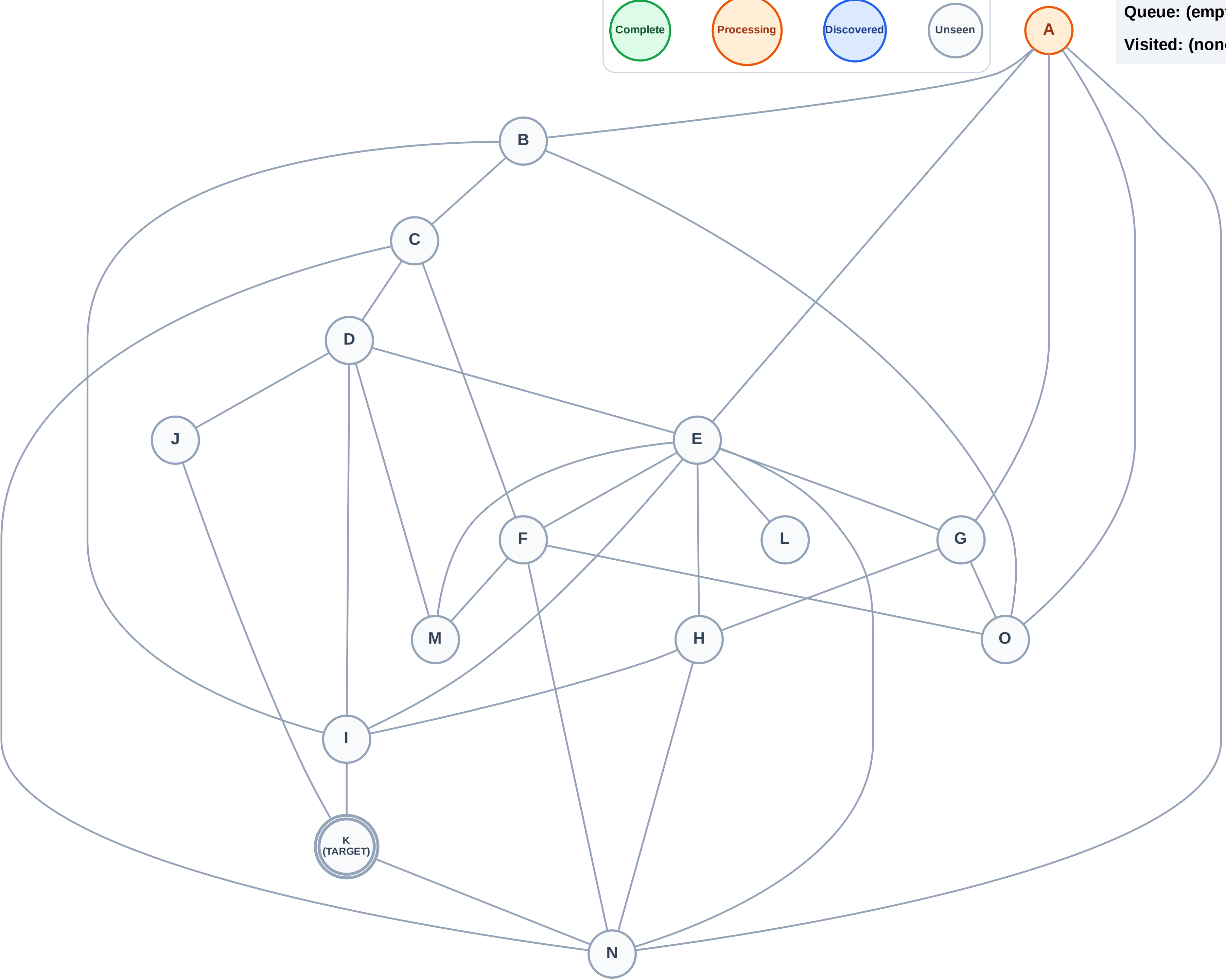
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

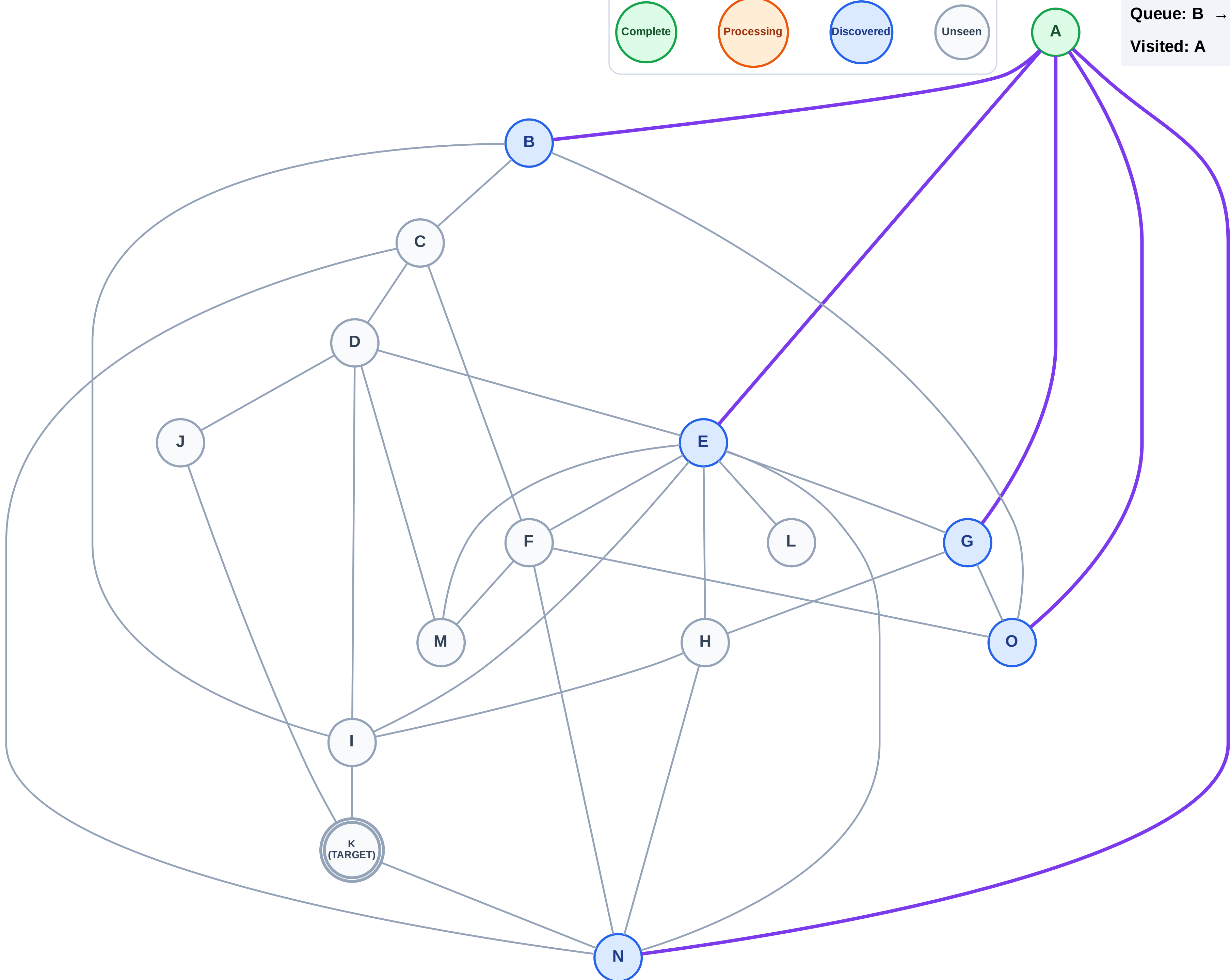
Step 3: Discover B, E, G, N, O from A

Legend



Queue: B → E → G → N → O

Visited: A



BFS Traversal
Step 4: Process node B

Legend

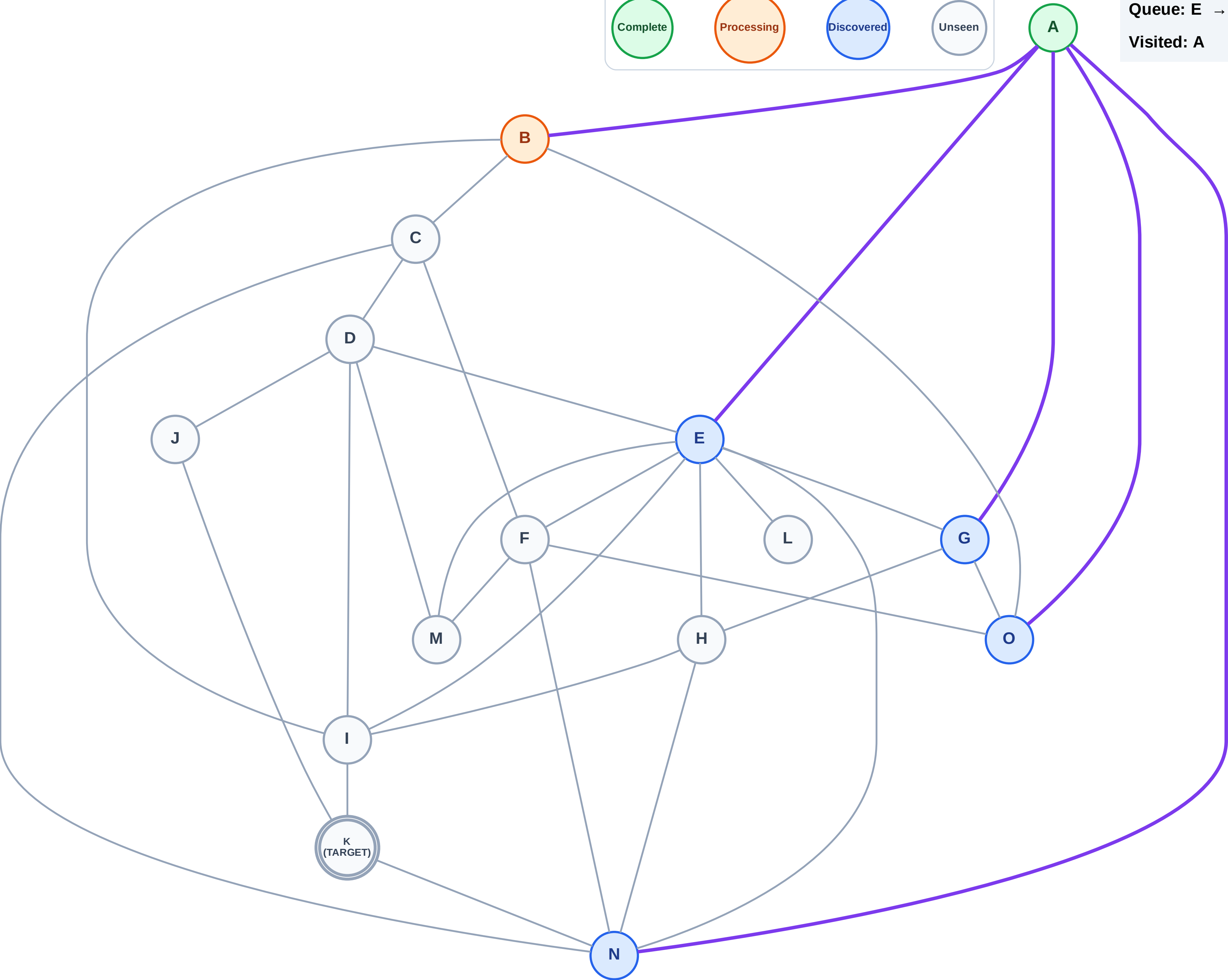
Complete

Processing

Discovered

Unseen

Queue: E → G → N → O
Visited: A



BFS Traversal

Step 5: Discover C, I from B

Legend



Complete



Processing

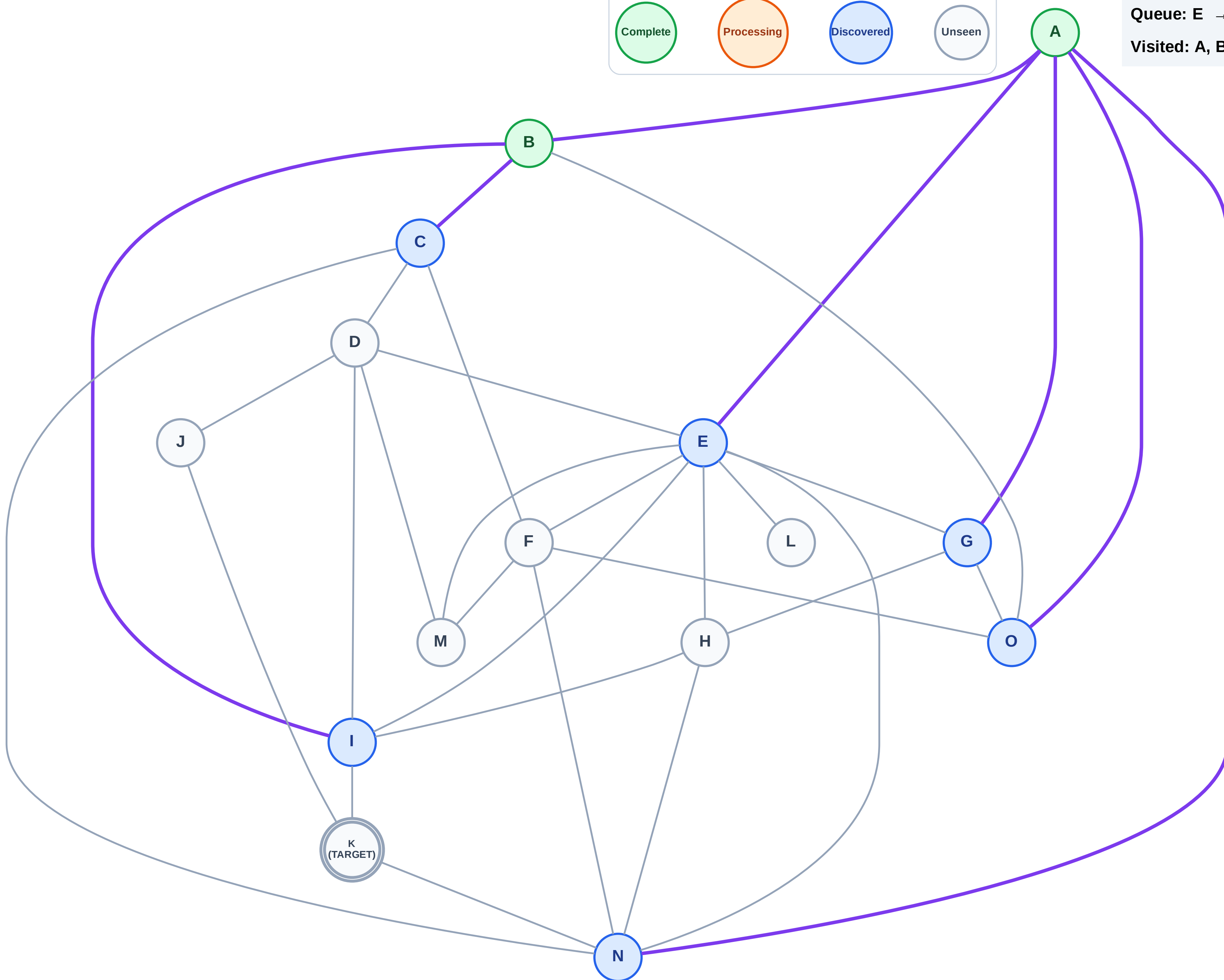


Discovered



Unseen

Queue: E → G → N → O → C → I
Visited: A, B



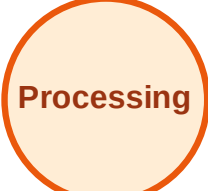
BFS Traversal

Step 6: Process node E

Legend



Complete



Processing



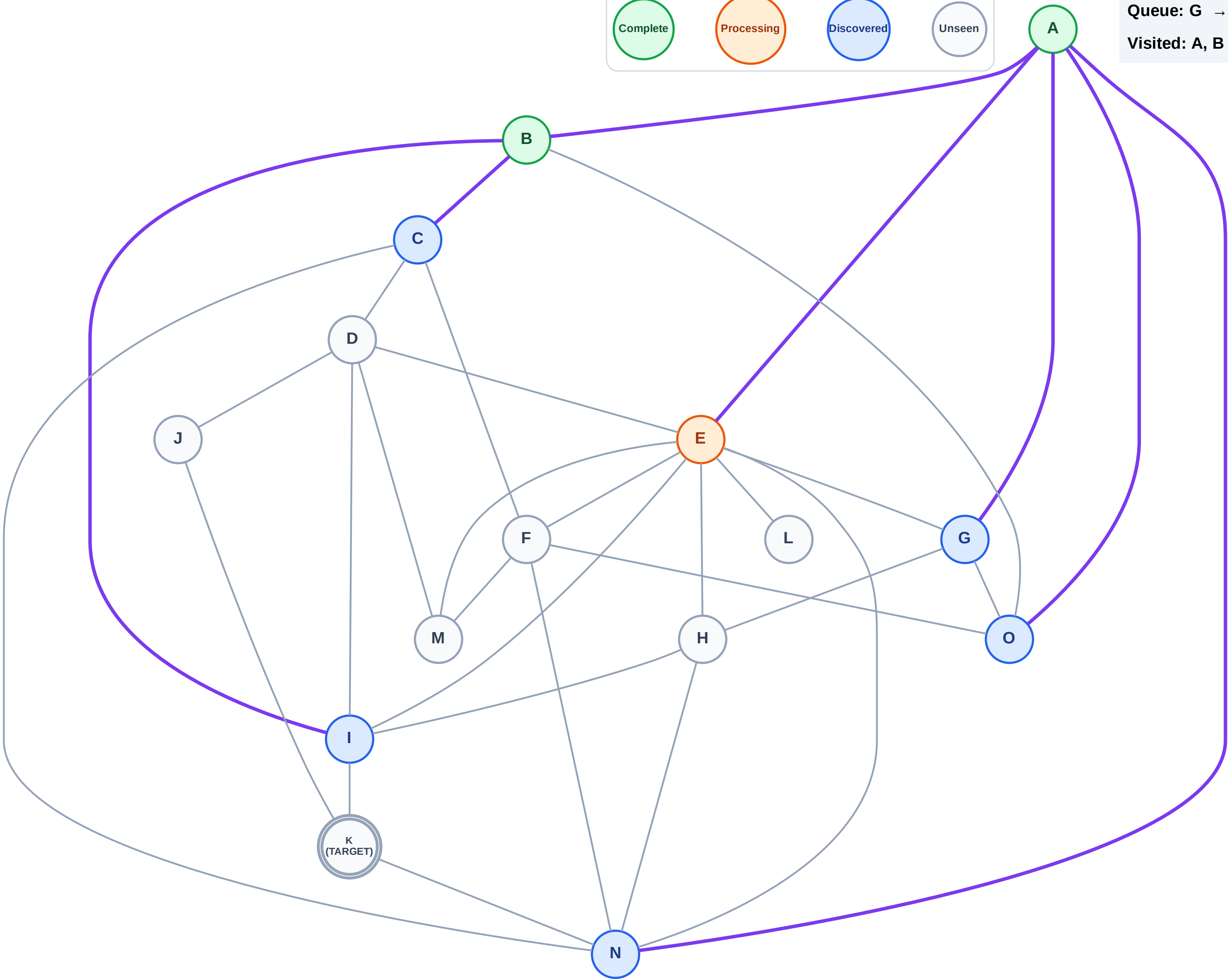
Discovered



Unseen

Queue: G → N → O → C → I

Visited: A, B



BFS Traversal

Step 7: Discover D, F, H, L, M from E

Legend



Complete



Processing



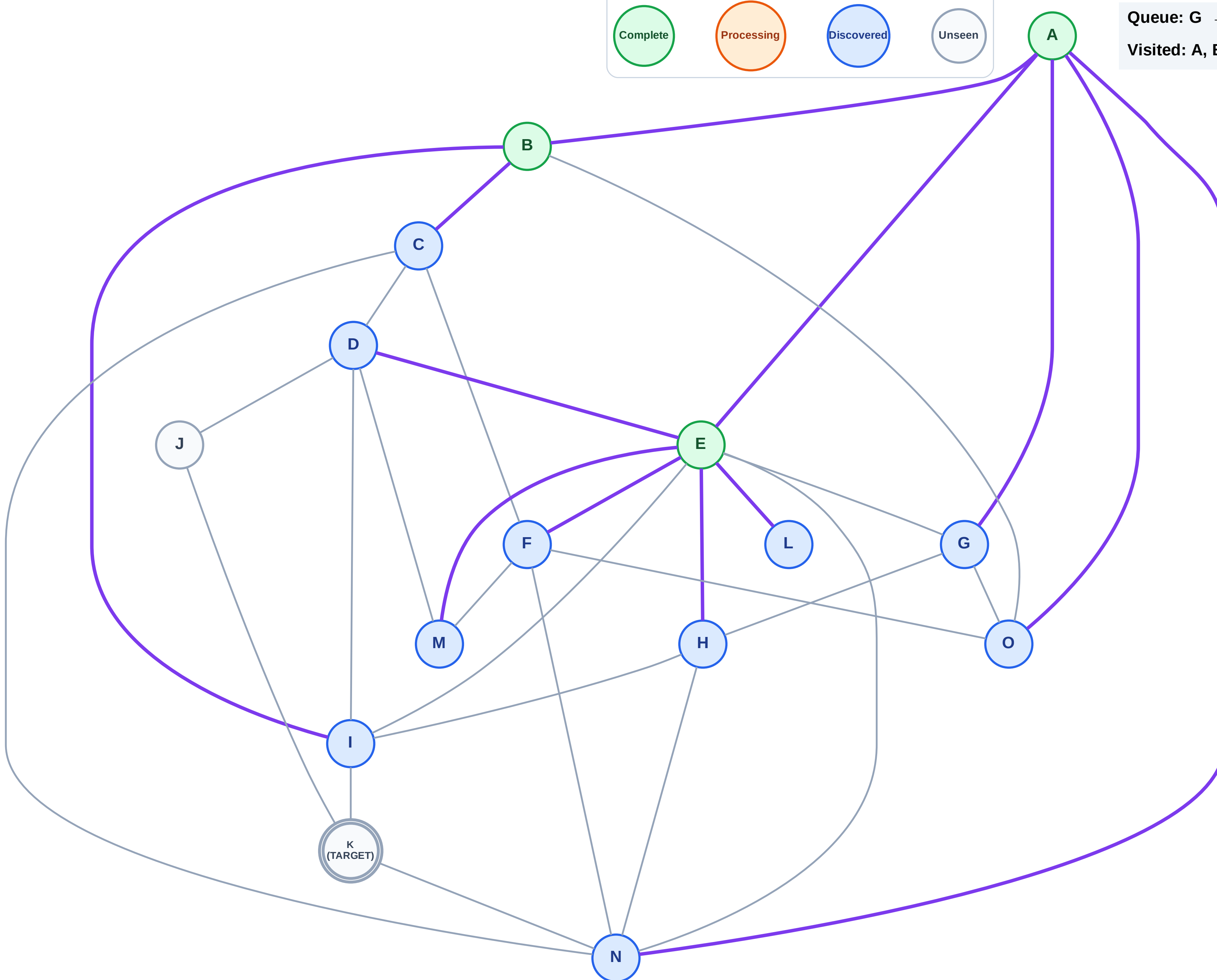
Discovered



Unseen

Queue: G → N → O → C → I → D → F → H → L → M

Visited: A, B, E



BFS Traversal
Step 8: Process node G

Legend

Complete

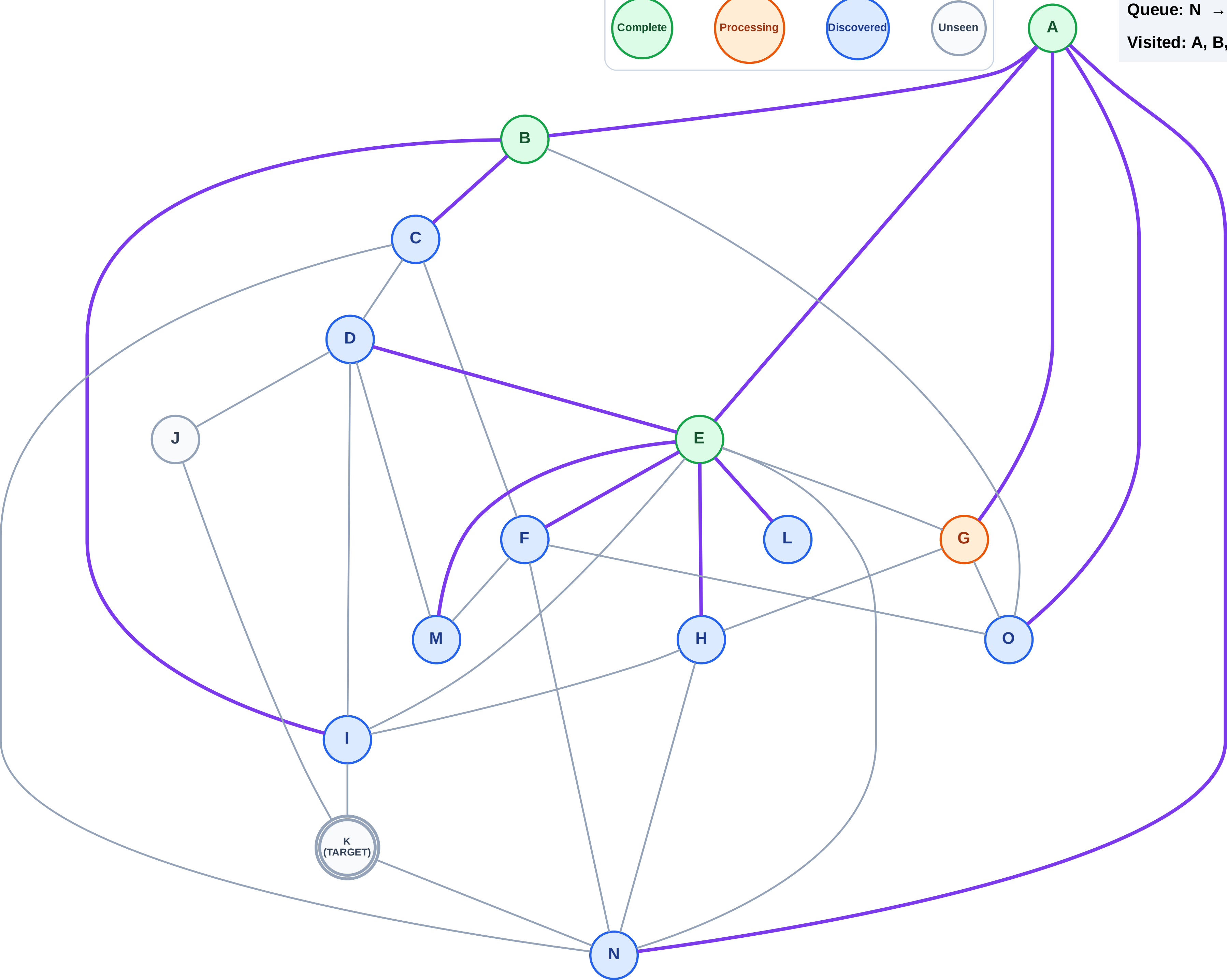
Processing

Discovered

Unseen

Queue: N → O → C → I → D → F → H → L → M

Visited: A, B, E



BFS Traversal

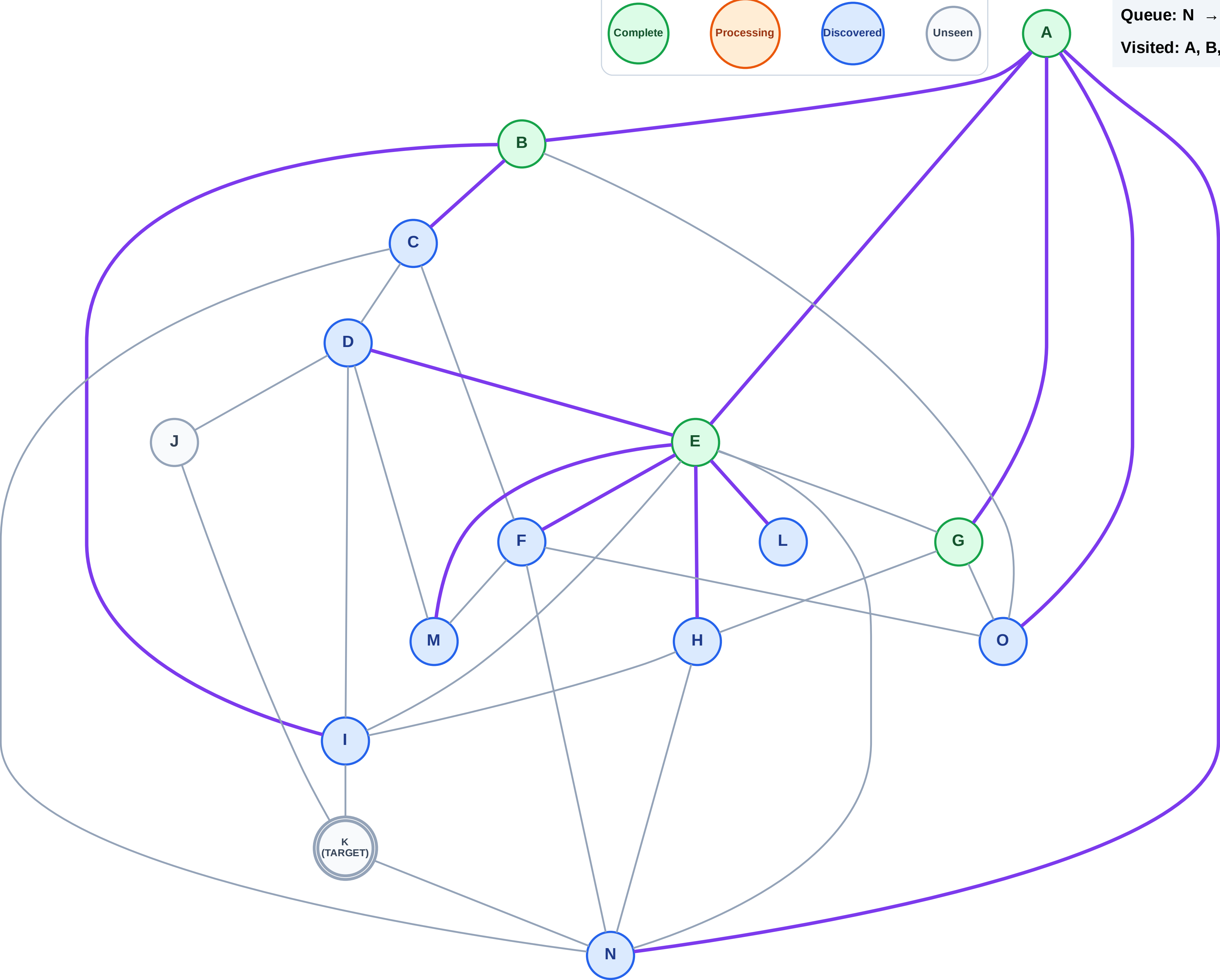
Step 9: No new neighbors from G

Legend



Queue: N → O → C → I → D → F → H → L → M

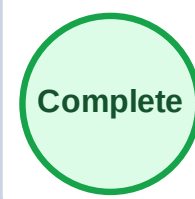
Visited: A, B, E, G



BFS Traversal

Step 10: Process node N

Legend



Complete



Processing

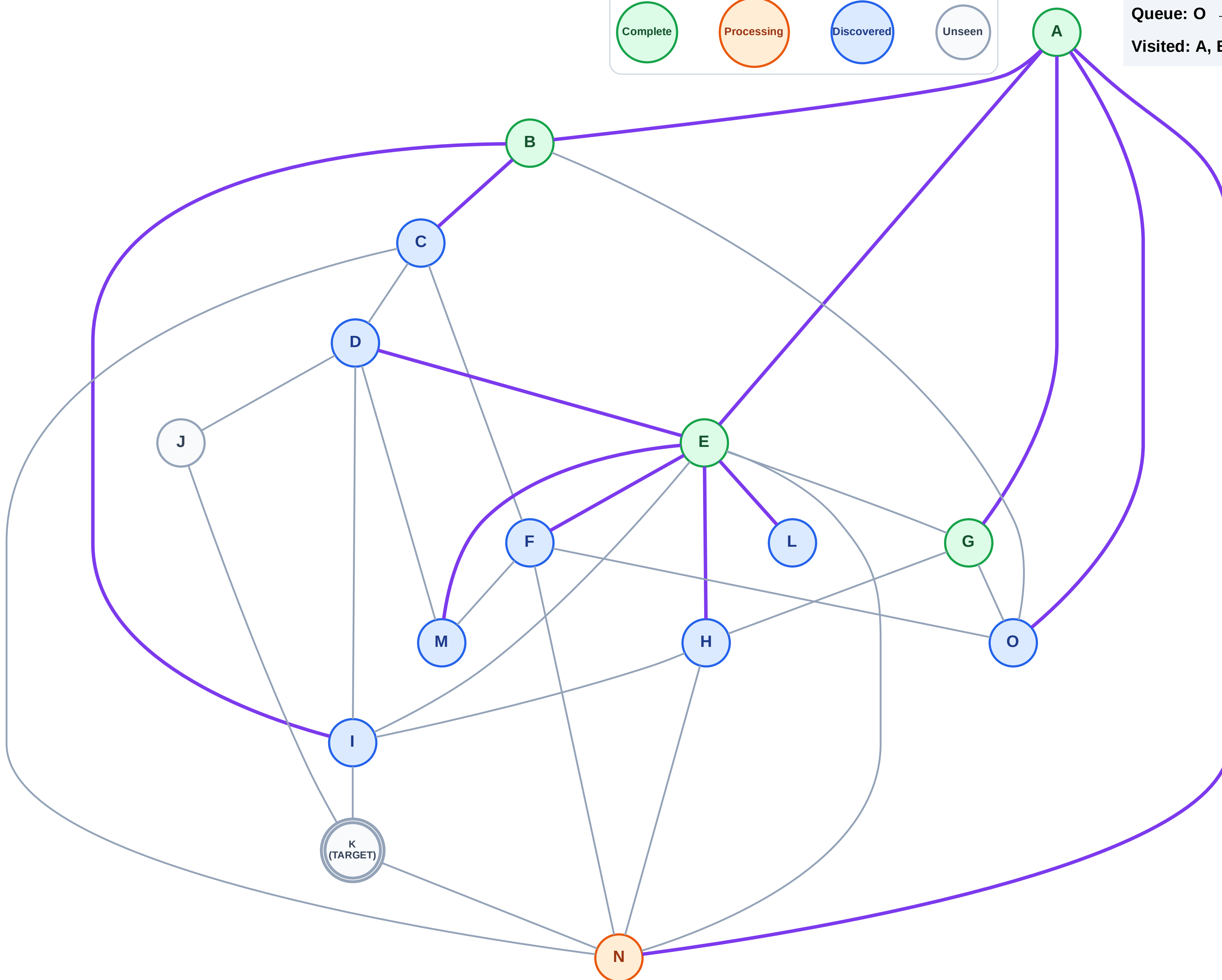


Discovered



Unseen

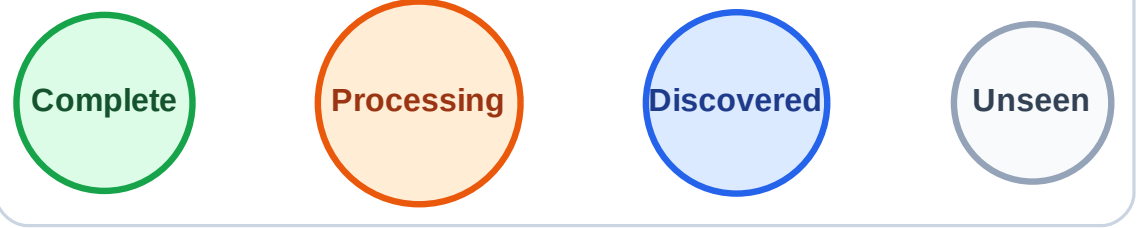
Queue: O → C → I → D → F → H → L → M
Visited: A, B, E, G



BFS Traversal

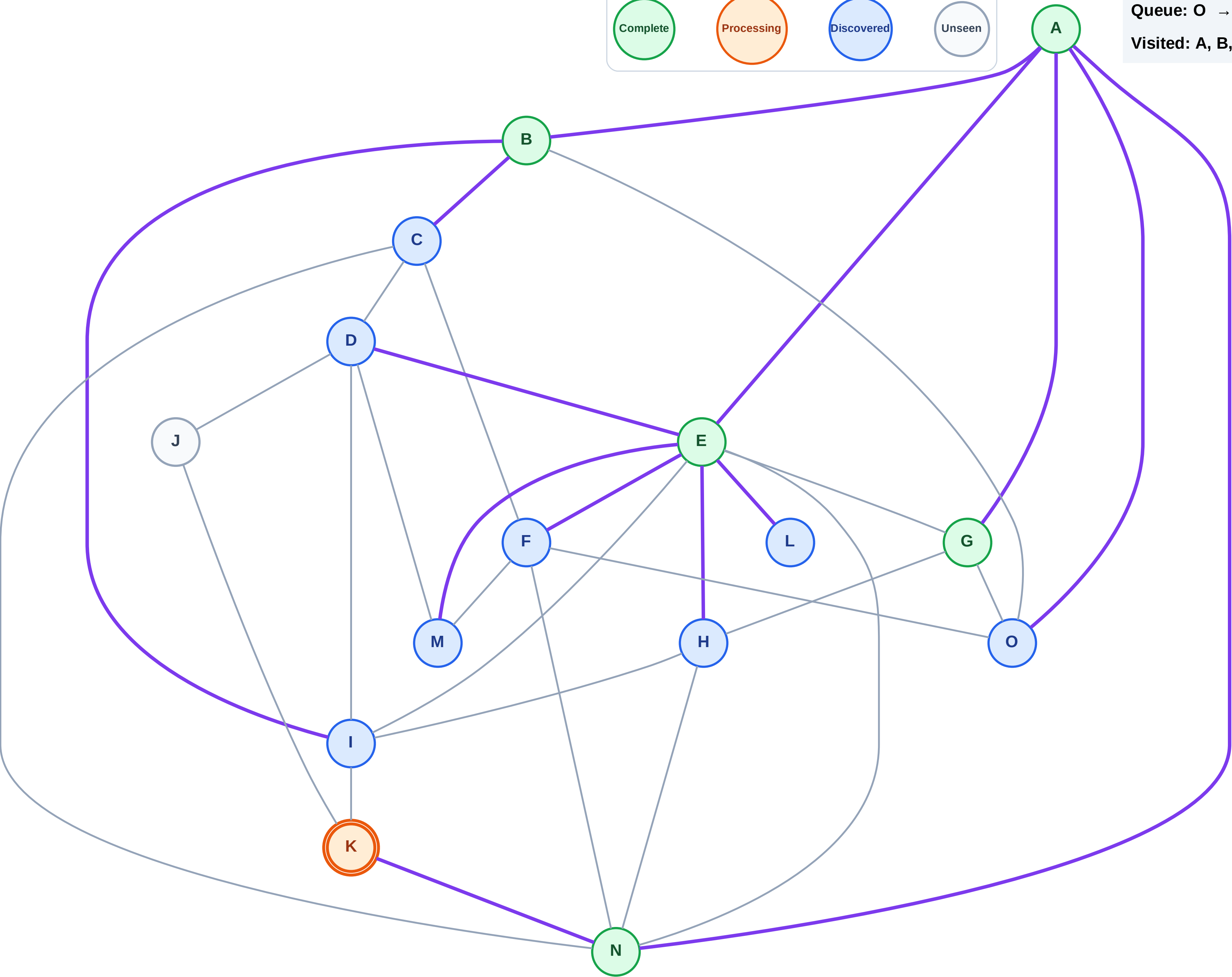
Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → N → K

Legend



Queue: O → C → I → D → F → H → L → M

Visited: A, B, E, G, K, N



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

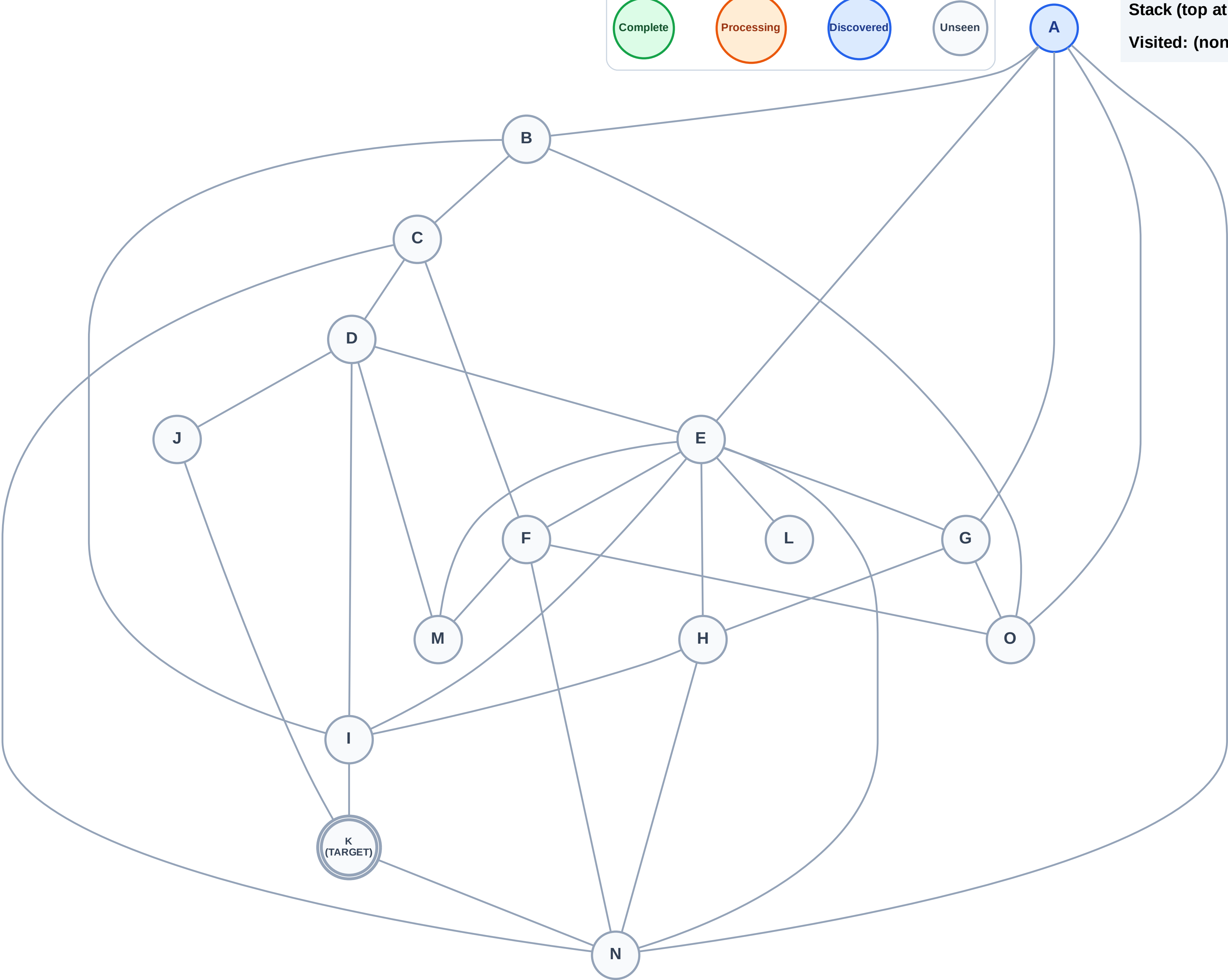
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

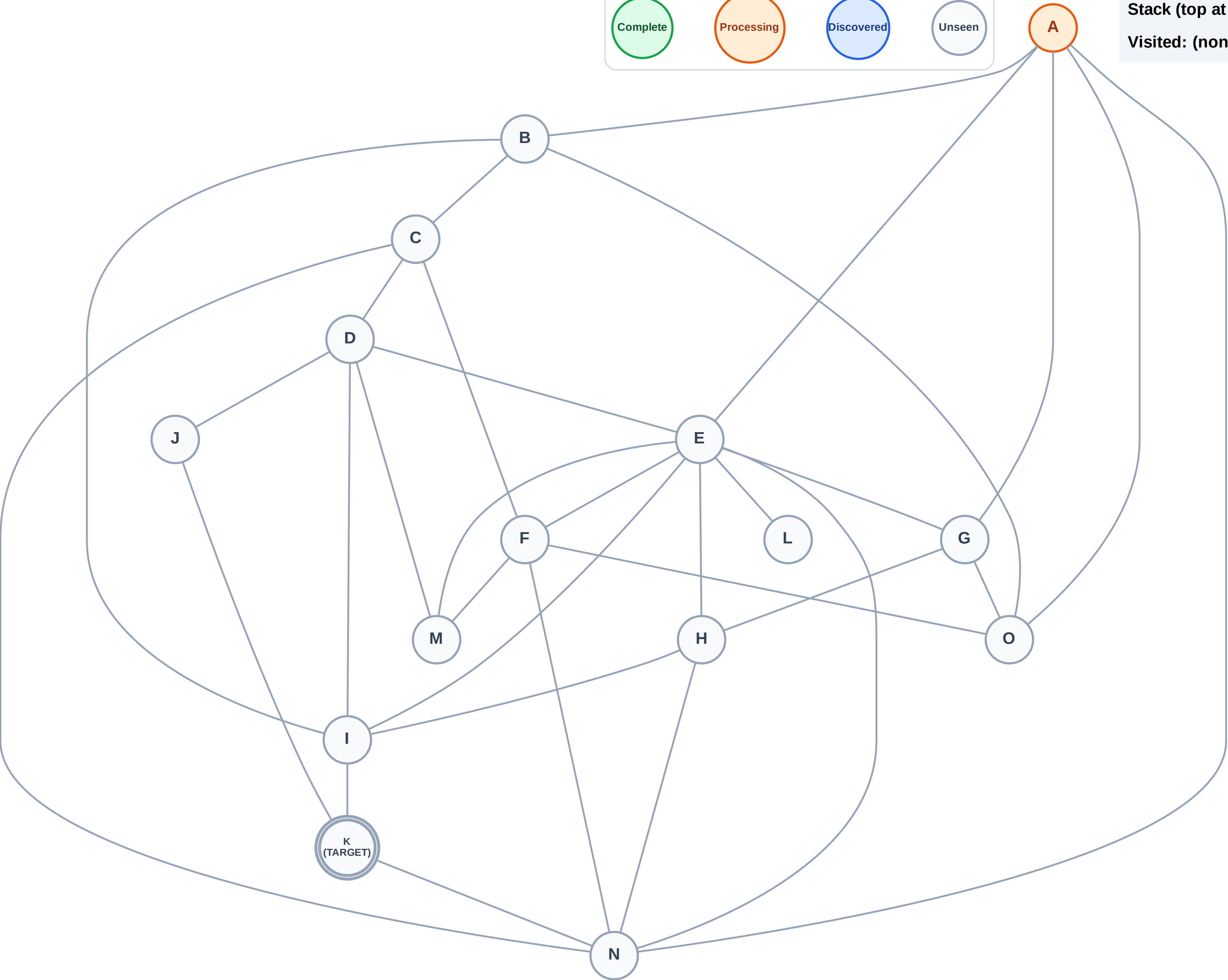
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

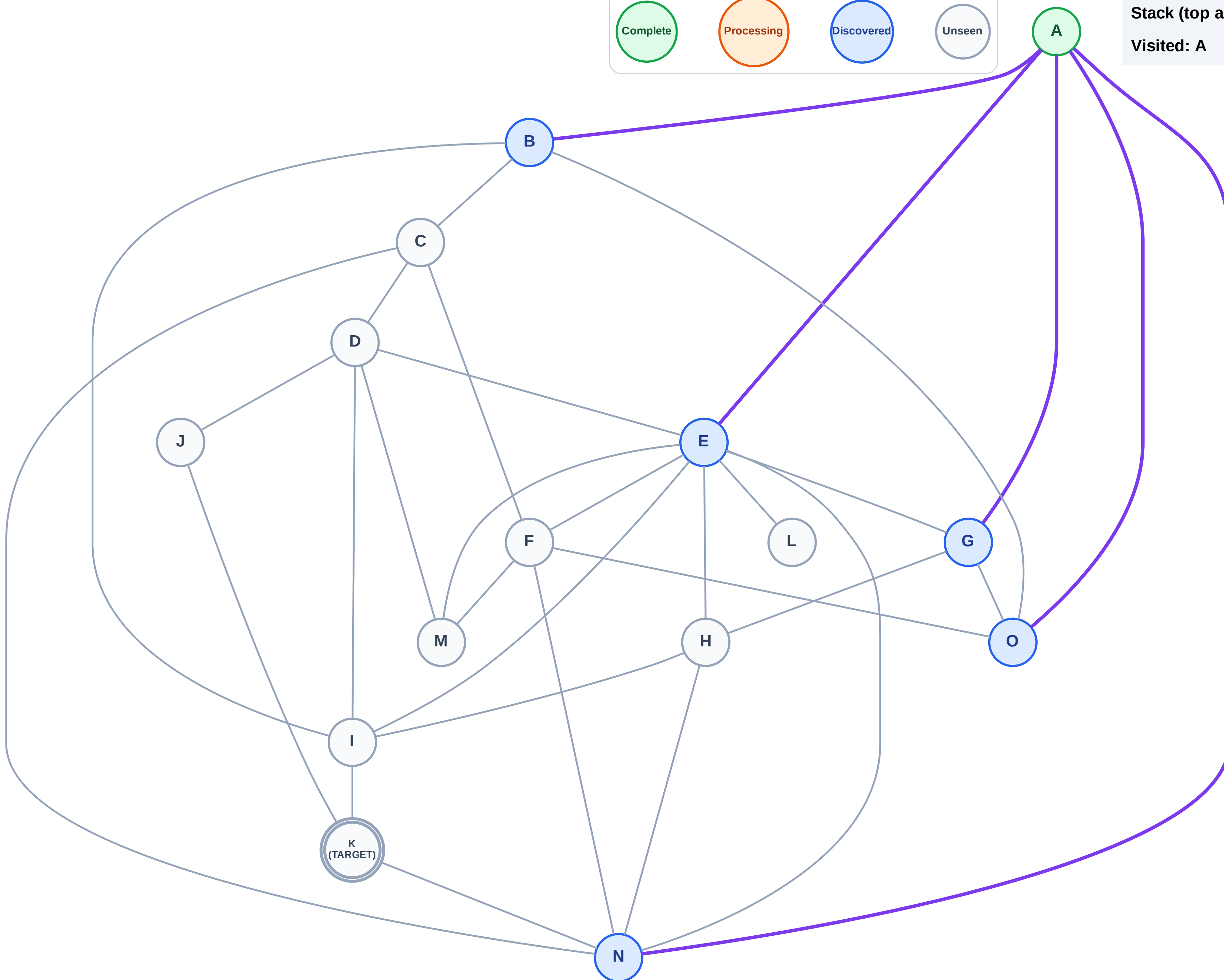
Step 3: Discover O, N, G, E, B from A

Legend



Stack (top at right): O | N | G | E | B

Visited: A



DFS Traversal

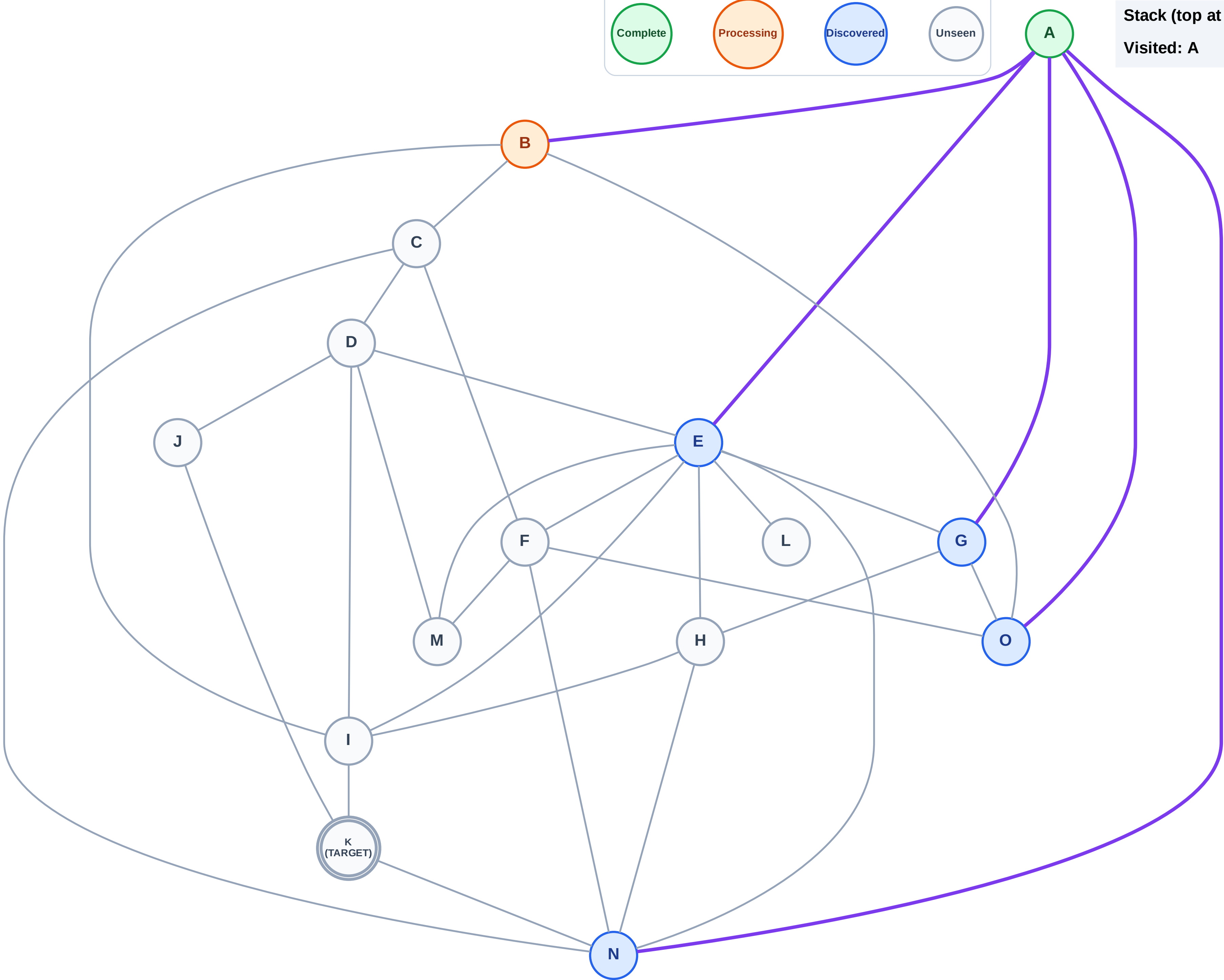
Step 4: Process node B

Legend



Stack (top at right): O | N | G | E

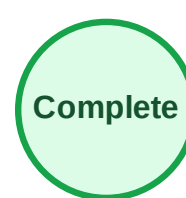
Visited: A



DFS Traversal

Step 5: Discover I, C from B

Legend



Complete



Processing



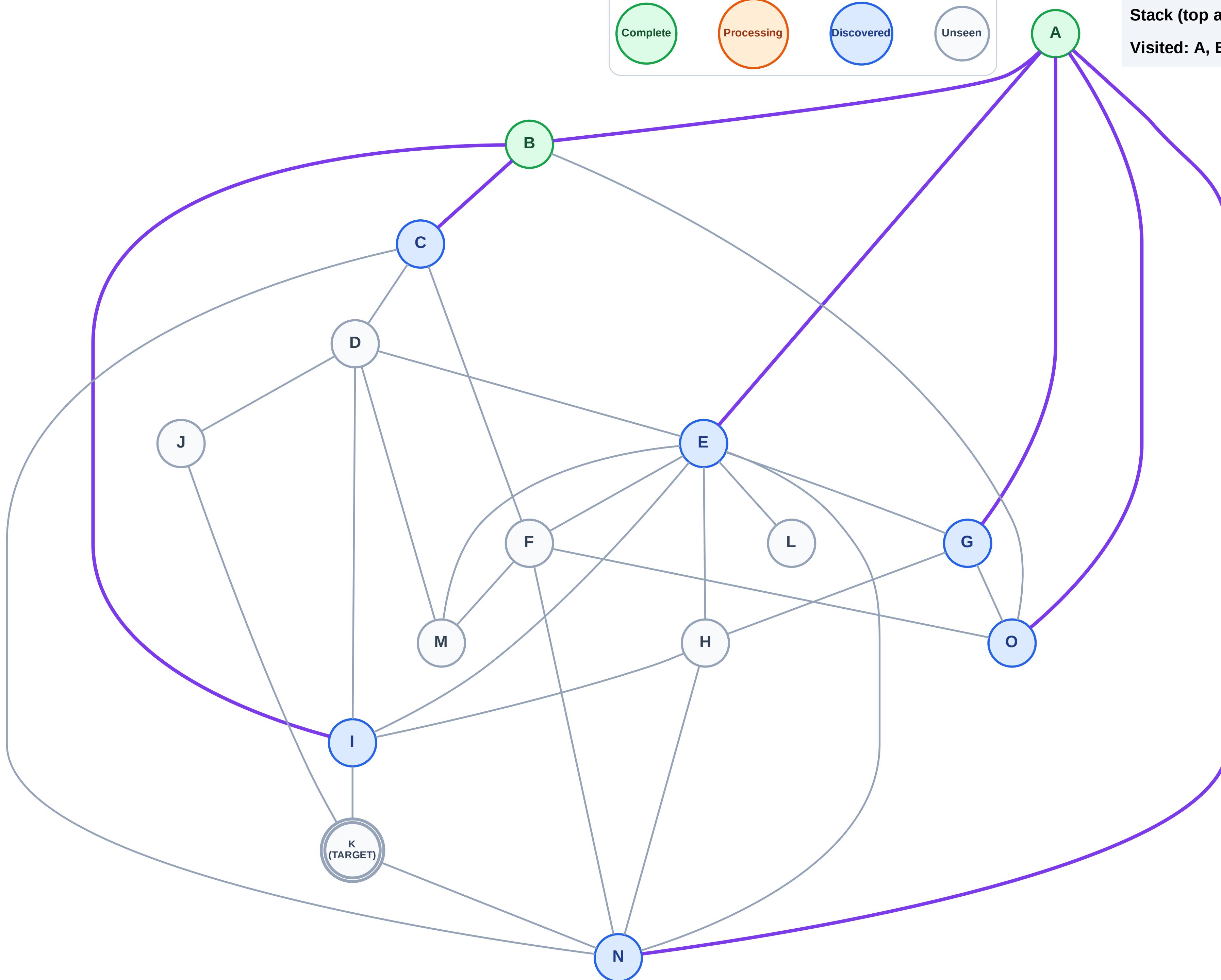
Discovered



Unseen

Stack (top at right): O | N | G | E | I | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

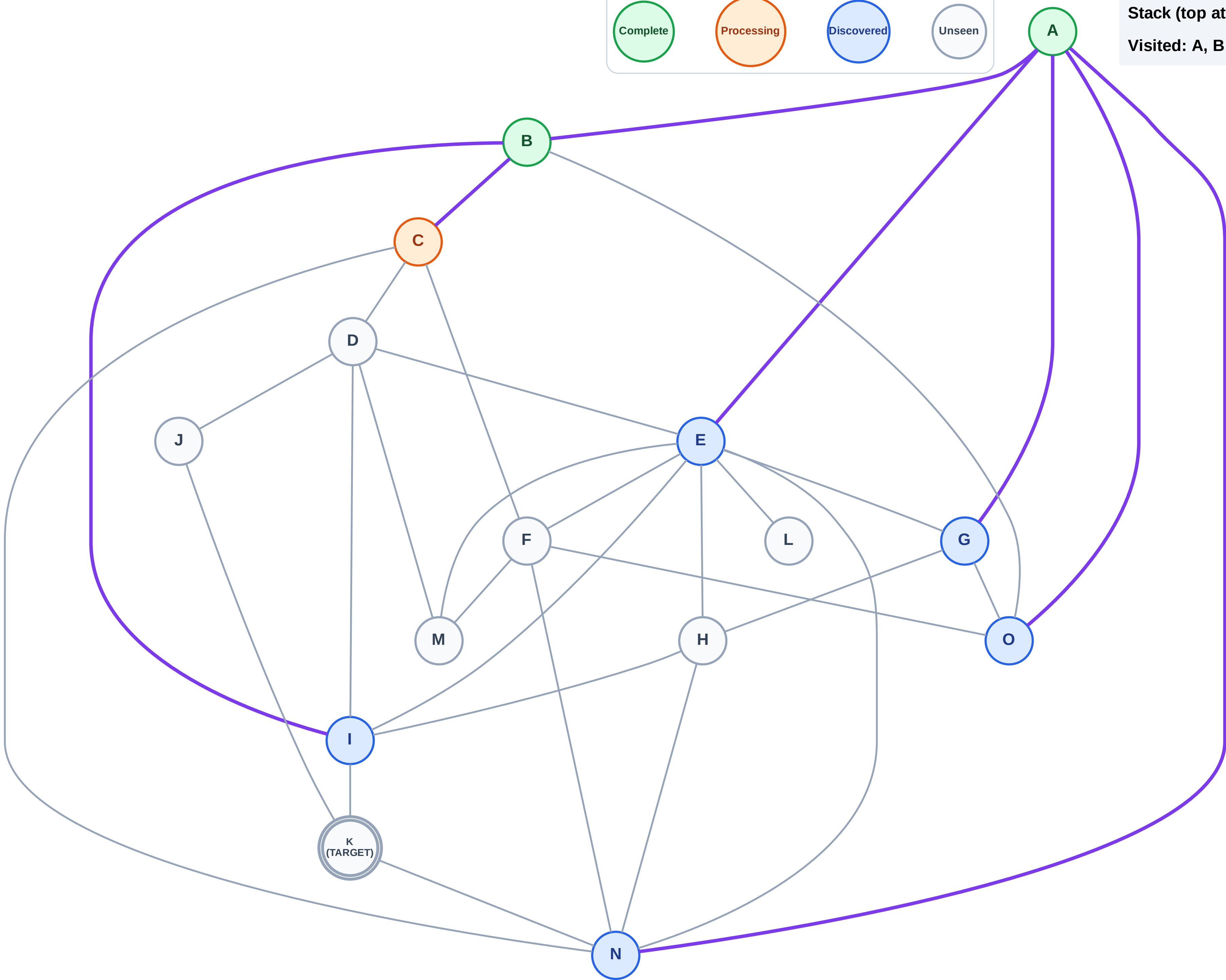
Processing

Discovered

Unseen

Stack (top at right): O | N | G | E | I

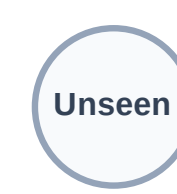
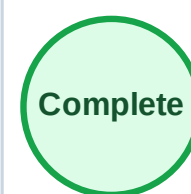
Visited: A, B



DFS Traversal

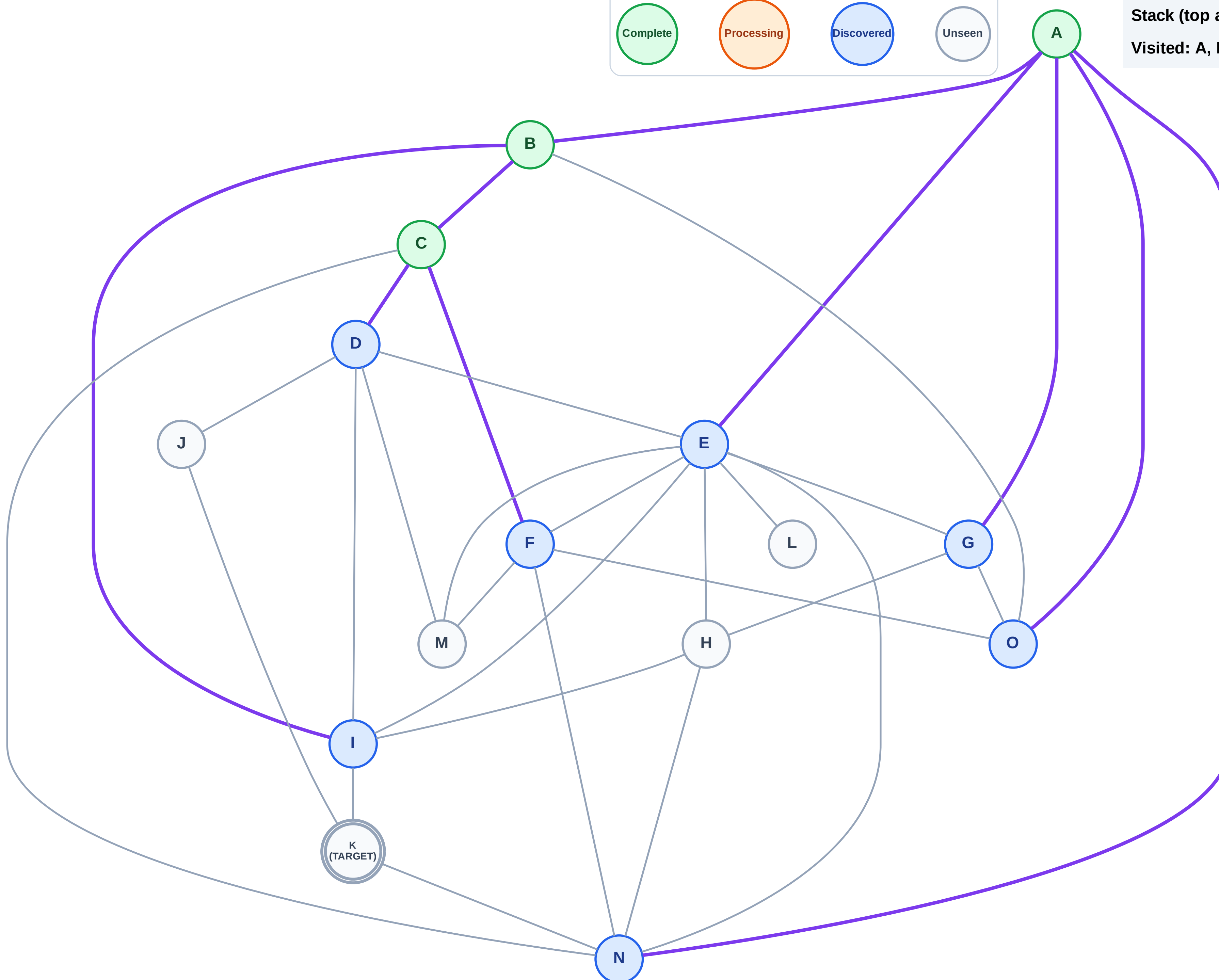
Step 7: Discover F, D from C

Legend



Stack (top at right): O | N | G | E | I | F | D

Visited: A, B, C



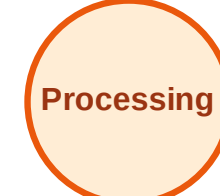
DFS Traversal

Step 8: Process node D

Legend



Complete



Processing



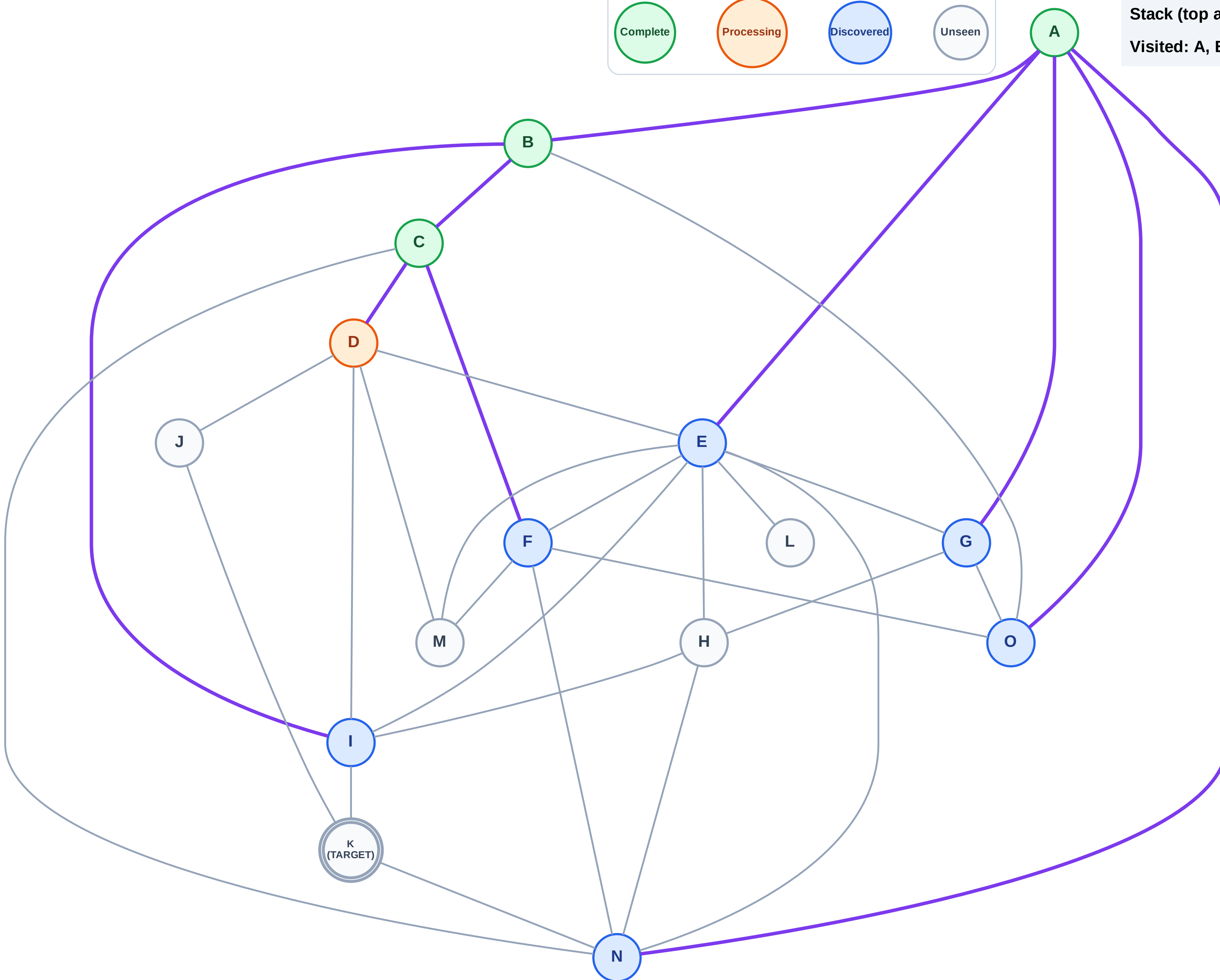
Discovered



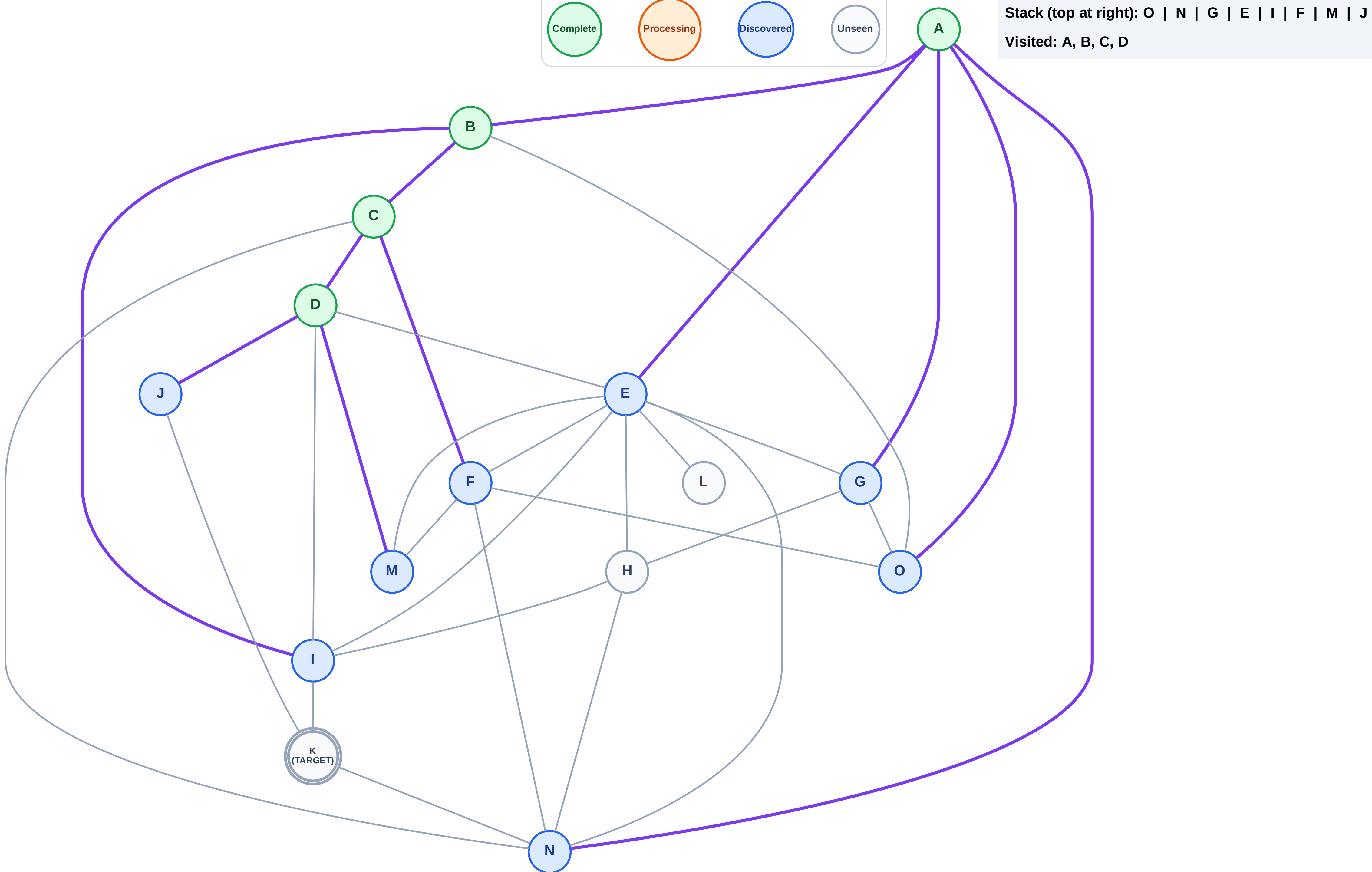
Unseen

Stack (top at right): O | N | G | E | I | F

Visited: A, B, C



Step 9: Discover M, J from



DFS Traversal

Step 10: Process node J

Legend

Complete

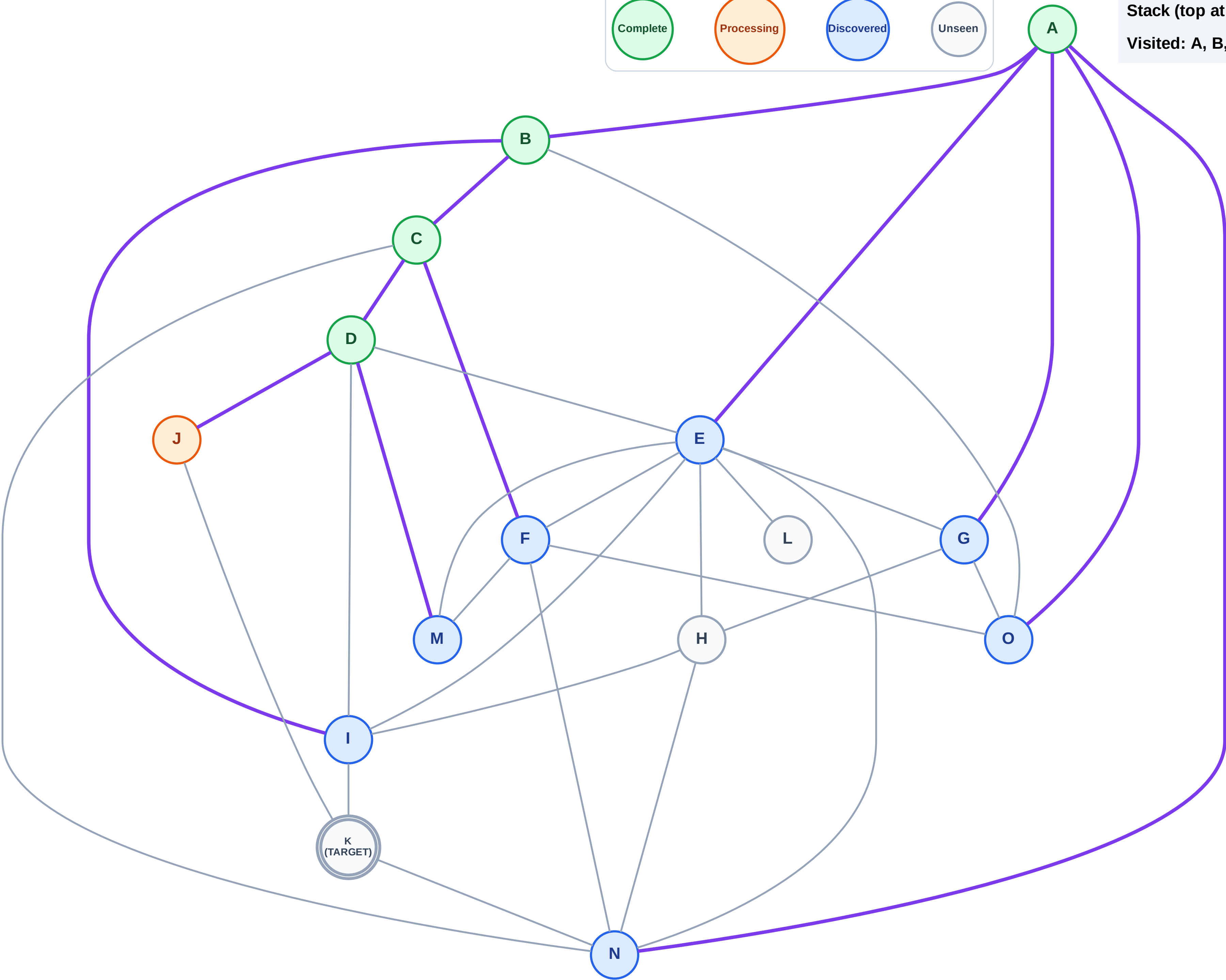
Processing

Discovered

Unseen

Stack (top at right): O | N | G | E | I | F | M

Visited: A, B, C, D



DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (5 hops): A → B → C → D → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | G | E | I | F | M

Visited: A, B, C, D, J, K

